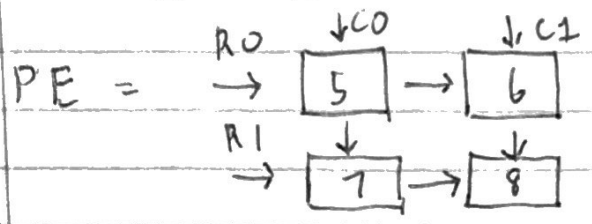
(1)

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \times \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$



(2)

Cycle	Input R0	Input R1	PE[0][0]ps	PE[0][1]ps
1	1		0 + (1x5) = 5	
2	3	2	0 + (3x5) = 15	0 + (1x6) = 6
3		4		0 + (3x6) = 18
4				

Cycle	PE[1][0]ps	PE[1][1]ps	Output C
1			
2	5 + (2x7) = 19		C[0][0] = 19
3	15 + (4x7) = 43	6 + (2x8) = 22	C[1][0] = 43, C[0][1] = 22
4		18 + (4x8) = 50	C[1][1] = 50

(3) a) 2 MACs x 4 outputs = 8 MAC operations

b) 2x Input values reused

c) 4 inputs + 4 weights + 4 outputs = 12 total accesses

(4) I believe the partial sum of the output matrix C would be fixed in the PE while Matrix A and B stream through.